# Gelareh Hasel Mehri

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## Summary


I have studied information security for my PhD, and I will graduate in Oct 2026. I am experienced with and enjoy working on data protection in the context of advanced access control systems. I have also designed and developed techniques for authentication anomaly detection. Our efforts have been recognized several times by the leading conference in the field, ACM SACMAT, and by the Microsoft Bounty Program.


## Education

**PhD in Information Security**, Eindhoven University of Technology (TU/e) 2022 – 2026

**Supervisor:** Dr. Nicola Zannone
**Area of expertise:** data protection through advancing access control in complex environments
• *Access decision-making in collaborative settings.* Modeled access decision making with game theory to accounts for multiple stakeholders' preferences; Investigated collaborative access control in online social networks and applied deep learning techniques for editing images while preserving fidelity and users' privacy expectations.
• *Anomaly detection to prevent access privilege misuse.* Applied machine learning methods to build profiles of user authentication behavior for Microsoft Entra ID (or Azure Active Directory) and medical sector. This involved advanced feature engineering and techniques to handle sparse historical access data.
• *Understandability and transparency in access control systems.* Identified the need for explaining access control decisions to involved parties, including administrators, stakeholders, and requesters. Developed frameworks: an approach based on causal reasoning to identify responsible policy elements, and a visualization framework for explaining XACML policy evaluation, implemented in SAFAX.

**M.Sc. in Communication Networks**, University of Tehran (UT)

**Thesis:** Orchestration of Routing and VNF Placement in 5G Networks
**Supervisor:** Dr. Vahid Shah-Mansouri
Modeled placement and routing of virtual network functions (VNFs) to minimize service and cloud provider costs simultaneously; proposed heuristics to tackle the NP-hard problem

## Skills

**Programming:** Python (IEEE certified), R, MATLAB (LinkedIn certified), C++
**Language:** English (TOEFL iBT 113/120), Dutch (A2 level), Farsi (Native)

## Experiences

**Modeling cloud authentication access behavior through user profiling for Microsoft Entra-ID** 2026

Developed user profiles to model authentication behavior using Microsoft Entra ID sign-in logs. Applied machine learning (i.e., clustering and classification) methods to capture access patterns and support anomaly detection. Context-aware feature engineering approaches were designed while addressing challenges related to limited historical data. Results were validated through sessions with experts at KEMBIT, a Microsoft partner in the Netherlands, confirming interpretable and informative insights into users' authentication behavior.
Publication: *Behavior-Driven User Profiling: A Case Study on Microsoft Entra ID Sign-In*, Journal of Computer Security (Sage Journals, 2026)

**Supporting comprehension of access control decisions through causality analysis** 2026

Proposed a framework to support explaining access control decisions by identifying the policy elements causing the outcomes. Introduced metrics to quantify the influence of these elements on access decisions. This work is part of the broader concept of explainable access control, which enables access policy analysis and helping stakeholders detect unintended behavior and guide systematic policy refinement.
Publications: *A Causal Framework for Explainable Access Control*, ACM Symposium on Access Control Models and

Technologies (SACMAT), Waterloo, Canada, 2026. and *Explaining Access Control Through Causal Reasoning*, International Conference on Information and Communications Security (ICICS), Japan, Fukui, 2026.

**Personalized privacy control using deep learning techniques for image localization and inpainting in online social networks** 2025

Proposed a privacy model for online social networks that enables users to conceal sensitive content in shared images based on personalized access policies, while preserving image fidelity. Implemented a pipeline that performs localization using deep learning models such as MaskDino and SAM, and inpainting with LaMa and SD. Conducted a user study to assess users' perceptions of compliance with privacy expectations.
Publication: *Personalized Privacy in OSNs: Evaluating Deep Learning Models for Context-Aware Image Editing*, Intelligent Systems with Applications (Elsevier), 2025

**Foundations for explainable access control** 2025

Conceptualized for the first time "Explainable Access Control" to address the need for better understanding access decisions. Introduced frameworks to support explaining access decisions.
Publication: *Towards Explainable Access Control*, ACM Symposium on Access Control Models and Technologies (SACMAT), New York, USA, 2025 (**Best Paper Award**)

**Intrusion detection opportunities in Microsoft Purview Unified audit logs** 2024

Analyzed detection capabilities of Microsoft Purview Unified audit logs for attacks on Exchange Online and SharePoint Online. Proposed an extended ATT&CK Matrix for Office 365 and log-based features for use in anomaly-based intrusion detection.
Publication: *Opportunities for Intrusion Detection in Cloud Productivity Environments*, TU Eindhoven, 2024 (**Microsoft Bounty Program Awarded**)

**Decision making for collaborative access control through game-theoretic modeling** 2024

Proposed a framework for access control in collaborative settings where multiple stakeholders manage shared resources. Developed a game-theoretic model that accounts for the privacy preferences of multiple parties and automates decision-making. The model ensures acceptability and optimality of collective access decisions without requiring user intervention. Explored various strategic environments (cooperative vs. non-cooperative bargaining) to represent distinct decision dynamics.
Publication: *A Bargaining-Game Framework for Multi-Party Access Control*, ACM Symposium on Access Control Models and Technologies (SACMAT), San Antonio, USA, 2024 (**Runner-up Best Paper**)

**Visualizing XACML policies and their evaluations to support comprehension** 2024

Created a visualization framework to support understanding of XACML policy evaluations and their contribution to access decisions. As part of explainable access control, enabled stakeholders to trace how decisions are derived and improved transparency and trust. Integrated the framework with SAFAX to visually represent policy evaluations and their contributions to access decisions.
Publication: *Enhancing the Comprehension of XACML Policies*, ACM Symposium on Access Control Models and Technologies (SACMAT), San Antonio, USA, 2024

**Anomaly detection for preventing privilege misuse in access control** 2023

Proposed a framework to complement access control with anomaly detection for real-time monitoring of access requests. Built user profiles using machine learning methods on historical access data from a Dutch hospital to detect deviations from expected authentication behavior and identify privilege misuse. This enables detection of harmful access patterns beyond traditional auditing approaches.
Publication: *Mitigating Privilege Misuse in Access Control through Anomaly Detection*, International Conference on Availability, Reliability and Security (ARES), Benevento, Italy, 2023

## References

- Nicola Zannone, *n.zannone@tue.nl*